



FFAR Final Progress Report

As part of closing out a FFAR grant, grantees must complete a Final Progress Report. Please use the template below to complete the programmatic report. The report must be submitted to FFAR within 90 days after the expiration or termination of a FFAR funded research grant. All questions about this form should be directed to grants@foundationfar.org.

The Final Progress Report communicates the cumulative results and major accomplishments of the funded grant research, including accomplishment for each aim and whether all research aims were fully completed. FFAR uses the content of the Final Progress Report to inform Congress, the Advisory Board, and other stakeholders on the successes, impacts and value of funding unique partnerships to support innovative science addressing today's food and agriculture challenges. If a question is not applicable to the project, please type "Nothing to report."

Grant Information	
Grant ID	EggTech-0000000028
Award Program	Egg Tech Prize Phase I
Project Title	Innovative techniques for fast in ovo sexing in poultry by detection of gender specific volatiles
Grant Start Date	01/01/2020
Grant End Date	06/30/2022
Total Grant Budget	\$ 573,907.22
Total FFAR Budget	\$ 277,732.22
Matching Funder(s)	KU Leuven, Belgium
Final Report Date	09/30/2022
Project Director/Principal Investigator Information	
Full Name	Dr. Jeroen Lammertyn
Email	Jeroen.lammertyn@kuleuven.be
Phone	321-632-1459
Grantee Organization Information	
Organization Name	KU Leuven
Address	Waaistraat 6, Box 5105
City, State Zip	3000, Leuven, Belgium
Tax ID	2834004300000
Authorized Signing Official (ASO) Information	
Name	Ms. Elke Lammertyn
ASO Title	Head European and International Projects
Email	Elke.lammertyn@kuleuven.be
Phone	321-632-0621



General Information

- 1.1. Please list the geographic location(s) – city, state, congressional district - where the work was conducted. If the work was conducted outside of the US, please list the city and country.**

Willem de Croylaan 42, box 2428, 3001 Leuven, Belgium

- 1.2. How many new jobs were created because of this grant funding?**

One job was created for a scientific researcher during 6 months in the first year.

- 1.3. How many jobs were maintained because of this grant funding?**

A total of two jobs for the key personnel and one job for the other personnel.

- 1.4. Have there been any changes to your organization's IRS 501(c)(3) non-profit status since you were awarded the grant? If yes, please explain.**

No

- 1.5. Has your organization undergone a recent name change? If so, please provide the new name of your organization.**

No



2. Scientific Report

2.1. Public Abstract (up to 500 words): The abstract should be 'stand-alone' and is intended for a general audience. It should be a concise overview/summary of the importance of the project, the issues that the project addressed, and the key findings of the project. The abstract should also clearly state how the results of the project help to address important needs in U.S. food and agriculture systems.

This project addresses the issue of the culling of day-old male chicks in the laying hen industry using *in ovo* sexing. Hereby, the sex of the embryo is determined during incubation. Subsequently, males can be euthanized at an earlier stage and incubation costs are decreased. One promising *in ovo* sexing approach is by measuring volatile organic compounds (VOCs, i.e., gases or scent) from chicken eggs during incubation. Preliminary results and published research demonstrated the link between VOCs and the embryo's sex. However, more research is required to prove the robustness and define its applicability in commercial settings.

Therefore, the **goal** of this project is to develop an approach for robustly measuring egg VOCs for sexing. Addressed issues are 1) accurately measuring egg VOCs, 2) increasing the number of measurements and 3) performing advanced data analysis techniques to robustly validate the technology. First, 1) eggs emit low concentrations of VOCs and it is required to limit the background signal. Therefore, the container in which the egg is positioned is of high importance. Materials such as Teflon bags and glass jars were defined as suitable to accumulate the egg's headspace. Next, headspace measurements were performed using a) sorbent materials such as silicon-coated stir bars or fibers with subsequent analysis on a gas chromatography-mass spectrometer (GC-MS) and b) direct headspace analysis using instruments such as selected ion flow tube mass spectrometry (SIFT-MS) and proton transfer reaction mass spectrometry (PTR-MS). Second, 2) by applying the stir bars or the Teflon sampling bags, the number of measurements could be increased. Third, 3) advanced modeling techniques and proper validation strategies were applied to have a good view on the practical robustness of the system.

Next **key findings** were obtained: 1) Egg VOCs fluctuate and originate from the environment, the degradation of egg contents, or the metabolism of the embryo. 2) Differences between fertilized eggs and unfertilized eggs became visible after 6 days due to embryonic metabolism. 3) When building sexing models per day, it was not possible to obtain prediction accuracies higher than 76 % between days 0 to 12. Finally, the focus was set on day 10, measuring 100 eggs using the three different VOC analysis techniques (GC-MS, SIFT-MS & PTR-MS). Similarly, it was not possible to perform sexing on the external validation set. Nevertheless, it was observed that, during multiple days, males emitted higher relative abundances of specific aldehydes, whereas females emitted higher relative abundances of specific alcohols. This was also partly verified by reported results in the literature. Toward future perspectives, it was concluded that this longitudinal information might be used to improve the accuracy of sex-determination models.



Generally, the results demonstrated that VOC *in ovo* sexing is not considered to be a highly accurate and robust approach in this stage to be applied in the US agricultural sector. Future research should highlight the further potential of using longitudinal information on egg VOCs to increase sex prediction accuracies. Finally, it is believed that the developed VOC measurement techniques are broader applicable in understanding the emission of VOC biomarkers of living organisms (i.e., plants and animals) in the US agricultural systems.

2.2. Provide a description or interpretation of how the key findings of the project can be used by the agriculture community. It is important the PI identifies how these findings have been, or may be, adopted or adapted in U.S. agriculture and food systems. If the findings cannot yet be applied, this section should address how they can be used to guide future planning or decision-making (up to 1,000-wordlimit).

Concerning adoption in U.S. agriculture and food systems, it is considered that VOC *in ovo* sexing is not yet ready for industrial application. It is concluded that further research is required to optimize sexing models. More specifically, it is suggested that sexing accuracies might be improved when the information on the VOC abundance over multiple incubation days is combined. However, to guide future planning or decision-making, it has to be stated that different factors will influence the sexing accuracy such as the genetic and environmental differences between flocks. Furthermore, research already indicated that sex differences within a specific breed are already difficult to observe.

In case VOC *in ovo* sexing would reach desirable sexing accuracy for industrial application, hatcheries would require more equipment to capture VOCs from eggs directly or through accumulation on a sorbent material to subsequently analyze the profile or specific compounds with a fast and accurate acquisition system. Ideally, only a limited selection of compounds would be necessary to define the sex using a target scan. This would increase the sensitivity and sexing accuracy whilst lowering cost and increasing speed.

Finally, VOCs carry a wide potential of containing biomarkers for monitoring health, development, or identification of pathogens in other agricultural products.



2.3. Research Outcomes/Impact (up to 1,500-word limit). The primary goal here is to answer questions such as “How did this project lead to improvements in U.S. agriculture and food systems?” or “How can the findings of this study guide or make improvement in future investigations and research?” Outcomes should be explained and classified in one of the following ways:

a) potential outcomes, i.e., findings, results, or recommendations that could impact U.S. agriculture and food systems, if used;

From the performed research and issues that have been tackled during the project, a robust approach has been developed in which agricultural products (i.e., chicken eggs) can be non-invasively assessed for their emitted VOCs. This includes the full pipeline of efficiently accumulating VOCs through different systems, robustly analyzing them on different platforms and the data analysis part to get the most information out of these measurements. Regardless of whether VOC in ovo sexing based on this pipeline would be efficient enough for commercial application, it is believed that the developed measurement strategies and materials have the potential for broader application. More specifically, exploration can be performed for new applications using VOC analysis systems to identify for instance biomarkers for disease diagnostics in plants or animals or assess food quality (e.g., meat, fruits & vegetables).

b) intermediate outcomes, i.e., how the findings, results, or recommendations have been used by others to influence practices, legislation, research/product design, training and so forth; and

Nothing to report

c) c. end outcomes, i.e., how findings, results, or recommendations have contributed to documented reductions in work-related morbidity, mortality, and/or exposure.

Nothing to report



2.4. What were the goals/specific aims of the project? If the approved application lists milestones/target dates for important activities or phases for this reporting period, identify these milestones and dates, as well as show actual completion dates or the percentage of completion of milestone targets. (up to 1,500-word limit)

During the project, the specific goals have been adapted to the preliminary outcomes of the experiments. The original goals and the justification for their change are described in section 2.5. An overview of the adapted goals that were approved after the first and second-year reports are described hereunder together with their actual completion. These goals were also defined together with their objectives in the timeline of the second-year report. Hereunder, the different goals are defined together with their objectives. A visual representation over time is presented in Figure 1.

1.1) Screening VOCs during incubation using SIFT & GCMS

- a) Optimization protocol: completed automated sampling GC-MS + bag measurements on SIFT-MS.
- b) Performing on days 0 – 16: completed but performed only on days 0 to 11 because of hypoxia/hypercapnia
- c) stainless steel & TDU: completed using glass jars & FEP bags. SPME fibers have been used instead of Tenax tubes for the extraction of egg VOCs
- d) SIFT-MS negative ionization module: Positive ionization was finally preferred for VOCs. The budget was reallocated to VOC analysis supplies & travel and contractual costs for measurements on PTR-ToFMS.

1.2) Sex-specific volatiles identification using SPME

- a) data analysis: completed with future perspectives. Higher sample numbers are required for higher robustness. Prior knowledge is needed to know the relevant day of incubation on which the chances are the highest to find sex-discriminating compounds.

1.3) VOCs emission patterns

- a) defining specific patterns of volatiles: completed, chemical classes are defined.
- b) interpreting patterns and linking them to chemical and biochemical processes described in the literature: completed, conference oral presentation performed and scientific paper being written. This paper is not published yet, because it is being enriched with the data from the PDMS-Twister measurements.

1.4) Comparison SIFT vs PTR

- a) Comparison experiment SIFT-MS vs. PTR-ToFMS: these were experiments conducted in function of future sexing experiments using consumption eggs. Two different case studies were performed wherein eggs from different management systems were compared and eggs that have been stored under different temperature conditions. In conclusion, the PTR had a slightly better performance in discriminating the different experiment groups from one another, although the improvement was not significantly better than the results of the SIFT-MS.



1.5) Twister optimization

a) looking for an optimized throughput and sensitive analysis approach: PDMS Twisters are the best-suited solution selected for identifying sex-specific compounds and allow for more robust models and reproducible results.

b) instrument and protocol optimizations: hardware & software issues delayed the setup and optimization of the external standards. However, this latter was not continued since it brought extra contamination to the measurements and influenced the egg VOC profile.

c) incubation experiments and comparison with SPME: The Twisters seemed to obtain a similar sensitivity level as the SPME and had a broader range of collected compounds. The throughput is at least doubled in comparison with SPME. In conclusion, the Twister will improve model construction and validation.

1.6) Twister fertile vs. infertile eggs

a) Identifying embryo-specific VOCs: This task has been completed and was important to distinguish the egg compounds that are originating from biochemical processes and not only chemical processes.

b) Look for emission patterns over time and quantify them: Completed, however, without an external standard for quantification since it was not beneficial and brought extra risks in contaminating the VOC profiles.

1.7) Incubation stages discrimination SIFT vs. PTR

a) obtaining conclusive results on the most optimal VOC analysis system for hatching egg volatiles: These experiments have been performed but the data analysis is not complete yet, since the effort was first attributed to section 1.8 since this was laying in the critical path.

1.8) Sex-specific volatiles identification using Twister/SIFT/PTR

a) PDMS Twister experiment for determining sex differences: this has been completed on the even incubation days between days 0 and 12. A follow-up of VOCs was made over these days and day 10 was selected to perform a big sexing experiment using 100 eggs and the Twister-GCMS.

b) SIFT or PTR to enlarge the dataset up to 100 – 300 eggs and focus on the identified compounds of the Twister experiment: Here, also day 10 was selected to perform the big sexing experiment on datasets of 100 eggs on both SIFT and PTR.

2.0) Glue selection & adsorbent identification setup preparation

Initiated work on the construction of a VOC accumulator and the identification of adsorbents.

2.1) Construction of volatile accumulator

a) Material selection: partly completed, potentially suitable adhesive has been selected and glass is the current material being used for the VOC accumulator.

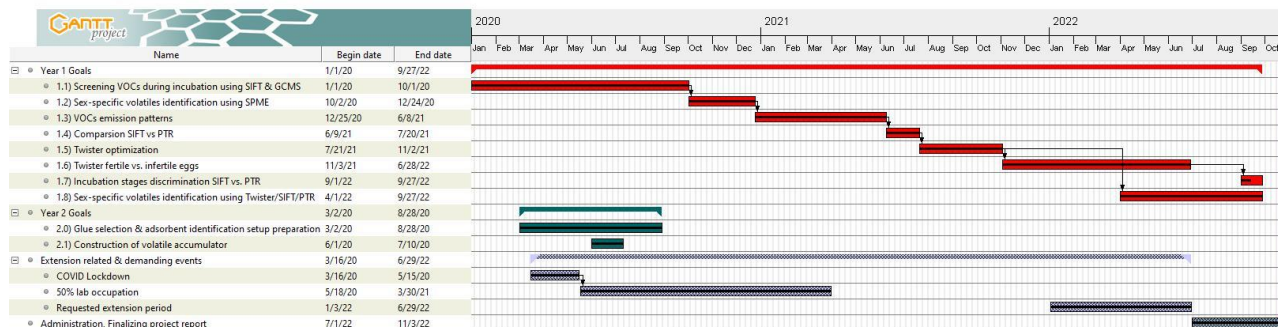


Figure 1: Gantt chart of the EggTech-0000000028 project containing all the specific goals

2.5. Have any of the major goals/specific aims or milestones for the project changed since the award? If so, please list the goal(s) that have changed and provide justification for the change from the approved goals. (up to 500-word limit)

Originally, the project goals were divided over two years.

In **year 1**, the focus was set on the development of a VOC sexing approach using GC- and SIFT-MS including the following three goals:

- 1.1) Screening egg volatiles during incubation using SIFT- & GC-MS,
- 1.2) Gender-specific volatiles identification
- 1.3) Large screening/sexing experiment using SIFT-MS.

In **year 2**, a chemical sensor would be developed that would indicate the sex of the embryo by a color change. The four defined goals were the following:

- 2.1) Volatile accumulator construction
- 2.2) Volatile accumulator readout using SIFT-MS
- 2.3) Adsorbent identification
- 2.4) Validation experiment with adsorbents inside the accumulator.

Generally, the development of innovative and fast techniques for poultry *in ovo* sexing remained the same throughout the whole project. However, a **major change** was a shift in the **scope of the second-year goals** where a chemical sensor would be developed with adsorbents specifically for indicating the embryo's sex through means of a color indication. This change of scope was made to focus the attention on having a better understanding of the identification, emission patterns, and origin of the egg VOCs (i.e., elaborating the year 1 goals). The potential of the chemical sensor is the detection of sex-specific VOCs without the need for expensive equipment and sampling. However, it is less sensitive than a GC-, SIFT- or PTR-MS and required first the identification of a sex-specific compound for which it would subsequently be developed.

Furthermore, **minor changes** in the scope have been applied to improve the robustness of the research. More specifically, PDMS Twisters have been used to replace the TD tubes to allow for the most optimal combination of higher throughput and sensitivity. The Twisters allowed increasing in the number of measurements on the white box approach of the GC-MS. This is in contrast to the black-box approach of the SIFT- and the PTR-MS that also allows for a higher sample number but does not have the identification of the compounds as on the GC-MS. Another minor change was the focus on VOC differences between fertile and infertile eggs to help in discriminating the difference between embryo-specific (and more relevant gender VOCs) and egg-specific VOCs.

2.6. What was accomplished under the goals/specific aims or milestones for the project? Please describe in detail, 1) major activities; 2) specific objectives; 3) significant results, including major findings, developments, or conclusions (both positive and negative); and 4) key outcomes or other achievements. Include discussion of stated goals not met. The emphasis in reporting in this section should focus on accomplishments. In the response, emphasize the significance of the findings to the scientific field. This section can be as technical as the PI would like. (up to 5,000-word limit)

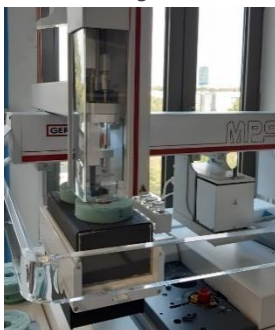
1.1) Screening VOCs during incubation using SIFT & GCMS

Here, the major objective was to optimize the SIFT- and GC-MS measurement protocol. A semi-automated incubation and extraction setup was created for the GC-MS. Eggs were placed in a custom-made 135 mL glass jar (Figure 2a) in a metal heating block on the GC-MS autosampler (Figure 2b). The glass jar had a screw cap with a silicon/Teflon septum. Through this septum, the SPME fiber was perforated. An optimized incubation-, extraction time, and oven program allowed a robust sampling and egg volatile analysis. The SPME headspace extraction with glass jars had a sufficiently low background signal. Moreover, the semi-automated setup of the SPME method facilitated tight scheduled and robust measurements. Subsequently, it was decided that more expensive stainless steel vessels were not needed.

a) SPME fiber & glass jars



b) Autosampler & metal heating block



c) FEP bag for SIFT-measurements

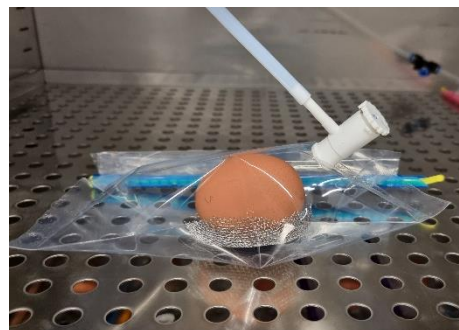


Figure 2: Recipients used for egg VOC measurements. a) Custom-made glass jars for the SPME fiber extraction with subsequent analysis on the GC-MS. b) Metal heating block on top of the GC-MS with automated SPME sampling of the glass jars by a robot arm. c) Egg positioned in an FEP Teflon bag filled with 300 mL filtered air for subsequent analysis on the SIFT-MS.

In parallel to the GC-MS experiment, the SIFT-MS measurement setup was optimized and consisted of incubating eggs separately in an FEP Teflon bag for approximately 2 hours with a subsequent analysis time of approximately 10 minutes. The bags were flushed with filtered air and filled with a volume of 300 mL (Figure 2c). These Teflon bags were inert and had a very low permeability. The advantage over rigid jars was the possibility of sampling the headspace without the need of flushing and subsequently diluting it. As such, the headspace concentration remained constant while the bag volume shrunk during the sampling.



1.2) Sex-specific volatiles identification using SPME

This section is described in more detail in the year 1 report and only the most important findings are highlighted hereunder. In short, a GC-MS volatile identification experiment was performed on egg incubation days 0 to 10. Given the long analysis time, a total of 8 eggs was measured per day. Due to the death of two embryos, only 2 males and 4 females were used to building the dataset. An extensive compound analysis was performed and resulted in 69 egg-originating compounds. Their relative abundance was calculated and the overall trends were studied over time. As a result, numerous VOCs could be matched with previous identification in the literature, including egg foreign compounds that might have been absorbed by the eggshell and that are slowly released during incubation.

Next to abundance, the component course over time was also studied. Here, it was observed that different groups of compounds had similar patterns over time. Acids were increasing over time, and compounds such as Pentanal/ Hexanal/ 1-Octanol and Benzaldehyde decreased in relative abundance. Other compounds first increased and then decreased, such as Phenol/ 1-Dodecanol and 1-Nonanol. In contrast, Octanal/ Nonanal and Decanal first decreased and later on increased in relative abundance. These different component trends demonstrate the compound dynamics over time and the relevance of identifying the most discriminating moment in time concerning sex. In the following sections, these dynamic patterns will be further studied using higher sample numbers.

Next to the GC-MS experiment, a SIFT-MS screening experiment was performed on egg incubation days 1, 2, 6, 8, 10, and 12. Here, a total of 33 eggs were measured. Infertile eggs and dead embryos were removed from the dataset and a total of 24 useful samples remained. The ratio was 9 males to 15 females. After applying the necessary preprocessing techniques, the classification efficiency of the partial least squares discriminant analysis (PLS-DA) models was depending on the type of validation strategy and was therefore inconclusive. For this reason, it was concluded that this dataset was too small for building a robust model. Another observation was the relatively low egg signal at limit sensitivity level of the SIFT-MS system.

In the search for higher sensitivity, contact was made with a PTR-ToFMS manufacturing company that demonstrated to us the higher sensitivity of their device (down to ppt levels) relative to our SIFT-MS (ppb levels). Furthermore, other advantages were the higher resolutions and the shorter scan times. As a result, we made contact with an academic partner in Antwerp (Belgium) to perform measurements on their PTR-ToFMS.



1.3) VOCs emission patterns

Generally, the focus was set on gaining deeper insights and knowledge about egg volatile organic compounds (VOCs). Given the fact that no dynamic pattern of egg volatiles during incubation has been described yet, it was decided to continue the reproduction of the profiling experiment in year 1 (section 1.2). This VOC profiling was still situated on the path toward sex identification since it gave an overview of all compounds emitted by eggs. When a relevant compound is identified for sex identification, it is beforehand known how its emission is changing over time and how it relates to other compounds. Moreover, it was relevant to know which day to focus on for conducting an extensive VOC sexing experiment.

First, the optimized setup and the analysis approach described in the year 1 report were used to perform this study. In contrast to year 1, the analysis approach was made more robust by applying a doublecheck identification with mass fragmentation spectra combined with retention indices calculated using an alkane standard. A total of 50 eggs was incubated and during each of the even days of incubation, a set of 7 eggs was measured (every day, a new subset of eggs). A total of 65 compounds was identified of which a majority was similar to the experiments in year 1. A principal components analysis (PCA; Figure 3a) and a partial least squares regression (PLSR) showed that a clear change in volatile profile was present from day 0 till day 12. Furthermore, a K-means cluster analysis divided the gases into 5 distinct clusters dominated by specific chemical classes: acids, aromatics, aldehydes, alcohols & ketones, and alkanes (Figure 3b). This subdivision into chemical classes approach is new in contrast to section 1.2. Interestingly, each of the clusters had a particular time-dependent pattern whereby the majority of the compounds decreased over time and the acids increased in their concentration towards day 12. A detailed overview of the dynamic patterns of these clusters together with the identified compounds and their chemical classes is presented in Table 1. Finally, the same eggs have also been measured on the SIFT. There, it was found that only the acids were abundant enough to be detected by the system and the GCMS was more sensitive.

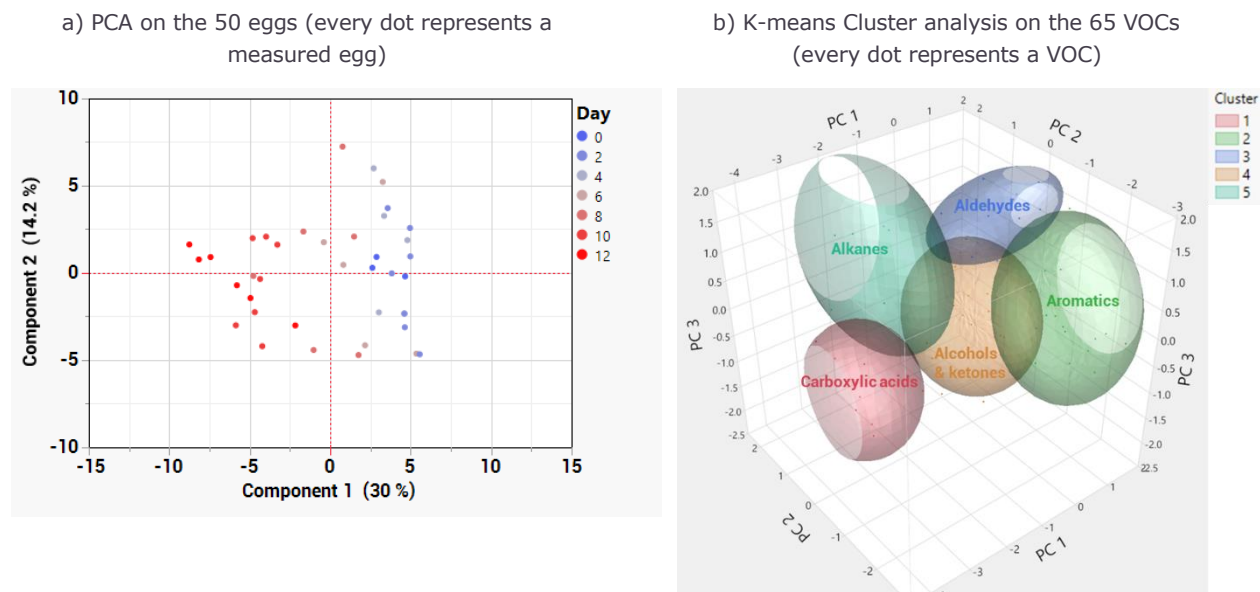


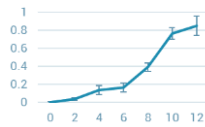
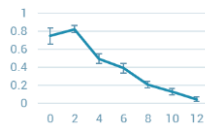
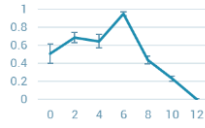
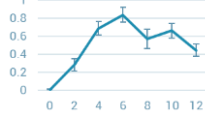
Figure 3: a) PCA on the eggs throughout 12 incubation days showing a time-dependent separation along the first principal component. b) K-means cluster analysis on the 65 components for their relative abundance course over time. A clear grouping was observed

Biochemically, it is known that fatty acids and amino acids are degraded into aldehydes, alcohols, ketones, and acids (Figure 4). A proportional increase of acids relative to the proportional decrease of alcohols, ketones, and aldehydes, after respectively 4 and 6 days, suggested that lower oxygen concentrations later in incubation might have resulted in a shift towards acid production. With respect to embryonic metabolism and gender differences, the cluster with alcohols & ketones deserves special attention. More specifically, earlier reported sex-specific compounds such as benzaldehyde, phenol, and 1-phenylethanone were identified in this cluster. Moreover, alcohols are reduced products from aldehydes and this is facilitated by alcohol dehydrogenase. This latter one has been demonstrated in the literature to be influenced by estradiol treatment (Hiremath *et al.*, 1992). Hypothetically, the formation of specific alcohols might be enhanced by steroid hormones during gonadal development from days 6.5 to 7.5 on. Following-up experiments need to confirm this (see section 1.8).



Figure 4: Summarized degradation pathways of Fatty- and amino acids (Belitz *et al.*, 2009)

Table 1: Overview of the 5 different clusters and their compounds

Carboxylic acids cluster  day	<table><tr><th>Iupac name</th><th>Chemical class</th></tr><tr><td>2-butoxyethanol</td><td>Alcohol</td></tr><tr><td>2,2-dimethylpropanoic acid</td><td>Carboxylic acid</td></tr><tr><td>2-methylpropanoic acid</td><td>Carboxylic acid</td></tr><tr><td>3-methylbutanoic acid</td><td>Carboxylic acid</td></tr><tr><td>Acetic acid</td><td>Carboxylic acid</td></tr><tr><td>Pentanoic acid</td><td>Carboxylic acid</td></tr><tr><td>Propanoic acid</td><td>Carboxylic acid</td></tr><tr><td>Penta-2,4-dienitrile</td><td>Nitrile</td></tr></table>	Iupac name	Chemical class	2-butoxyethanol	Alcohol	2,2-dimethylpropanoic acid	Carboxylic acid	2-methylpropanoic acid	Carboxylic acid	3-methylbutanoic acid	Carboxylic acid	Acetic acid	Carboxylic acid	Pentanoic acid	Carboxylic acid	Propanoic acid	Carboxylic acid	Penta-2,4-dienitrile	Nitrile	-) Carboxylic acids are oxidative degradation products of aldehydes (Belitz et al., 2009) -) It is assumed that the decrease of oxygen due to higher metabolism, leads to an increase in the production of carboxylic acids over time																						
Iupac name	Chemical class																																									
2-butoxyethanol	Alcohol																																									
2,2-dimethylpropanoic acid	Carboxylic acid																																									
2-methylpropanoic acid	Carboxylic acid																																									
3-methylbutanoic acid	Carboxylic acid																																									
Acetic acid	Carboxylic acid																																									
Pentanoic acid	Carboxylic acid																																									
Propanoic acid	Carboxylic acid																																									
Penta-2,4-dienitrile	Nitrile																																									
Aromatics cluster  day	<table><tr><th>Iupac name</th><th>Chemical class</th></tr><tr><td>pentan-1-ol</td><td>Alcohol</td></tr><tr><td>phenylmethanol</td><td>Alcohol</td></tr><tr><td>2-methylbutanal</td><td>Aldehyde</td></tr><tr><td>3-methylbutanal</td><td>Aldehyde</td></tr><tr><td>decanal</td><td>Aldehyde</td></tr><tr><td>dodecane</td><td>Alkane</td></tr><tr><td>octadecane</td><td>Alkane</td></tr><tr><td>2,6-di(propan-2-yl)naphthalene</td><td>Aromatic</td></tr><tr><td>styrene</td><td>Aromatic</td></tr><tr><td>toluene</td><td>Aromatic</td></tr><tr><td>2-decyloxirane</td><td>Epoxy</td></tr><tr><td>(3,3,5-trimethylcyclohexyl) 2-methylprop-2-enoate</td><td>Ester</td></tr><tr><td>2,3-diacetyloxypropyl acetate</td><td>Ester</td></tr><tr><td>cyclohex-2-en-1-one</td><td>Ketone</td></tr><tr><td>3-methylbut-2-enitrile</td><td>Nitrile</td></tr><tr><td>2,6-di-tert-butyl-4-methylphenol</td><td>Phenol</td></tr><tr><td>bis(2-methylpropyl) benzene-1,2-dicarboxylate</td><td>Phthalate ester</td></tr><tr><td>(methylidisulfonyl)methane</td><td>Sulfide</td></tr></table>	Iupac name	Chemical class	pentan-1-ol	Alcohol	phenylmethanol	Alcohol	2-methylbutanal	Aldehyde	3-methylbutanal	Aldehyde	decanal	Aldehyde	dodecane	Alkane	octadecane	Alkane	2,6-di(propan-2-yl)naphthalene	Aromatic	styrene	Aromatic	toluene	Aromatic	2-decyloxirane	Epoxy	(3,3,5-trimethylcyclohexyl) 2-methylprop-2-enoate	Ester	2,3-diacetyloxypropyl acetate	Ester	cyclohex-2-en-1-one	Ketone	3-methylbut-2-enitrile	Nitrile	2,6-di-tert-butyl-4-methylphenol	Phenol	bis(2-methylpropyl) benzene-1,2-dicarboxylate	Phthalate ester	(methylidisulfonyl)methane	Sulfide	-) Styrene and Toluene have been earlier identified as foreign compounds that might have been taken up by the egg and afterward again be re-emitted (Webster et al., 2015) -) Toluene can be metabolized into phenylmethanol -) Esters might originate from plant-specific products Amino acid Strecker degradation (Belitz et al., 2009) -) 2-methylbutanal might originate from Isoleucine -) 3-methylbutanal might originate from Leucine		
Iupac name	Chemical class																																									
pentan-1-ol	Alcohol																																									
phenylmethanol	Alcohol																																									
2-methylbutanal	Aldehyde																																									
3-methylbutanal	Aldehyde																																									
decanal	Aldehyde																																									
dodecane	Alkane																																									
octadecane	Alkane																																									
2,6-di(propan-2-yl)naphthalene	Aromatic																																									
styrene	Aromatic																																									
toluene	Aromatic																																									
2-decyloxirane	Epoxy																																									
(3,3,5-trimethylcyclohexyl) 2-methylprop-2-enoate	Ester																																									
2,3-diacetyloxypropyl acetate	Ester																																									
cyclohex-2-en-1-one	Ketone																																									
3-methylbut-2-enitrile	Nitrile																																									
2,6-di-tert-butyl-4-methylphenol	Phenol																																									
bis(2-methylpropyl) benzene-1,2-dicarboxylate	Phthalate ester																																									
(methylidisulfonyl)methane	Sulfide																																									
Aldehydes cluster  day	<table><tr><th>Iupac name</th><th>Chemical class</th></tr><tr><td>octan-1-ol</td><td>Alcohol</td></tr><tr><td>cyclohex-3-ene-1-carbaldehyde</td><td>Aldehyde</td></tr><tr><td>heptanal</td><td>Aldehyde</td></tr><tr><td>hexanal</td><td>Aldehyde</td></tr><tr><td>nonanal</td><td>Aldehyde</td></tr><tr><td>octanal</td><td>Aldehyde</td></tr><tr><td>pentanal</td><td>Aldehyde</td></tr><tr><td>undecanal</td><td>Aldehyde</td></tr><tr><td>4,6-dimethyldodecane</td><td>Alkane</td></tr><tr><td>3,3-dimethyl-2-benzofuran-1-one</td><td>Furanone</td></tr><tr><td>6-methylhept-5-en-2-one</td><td>Ketone</td></tr><tr><td>1,3-benzothiazole</td><td>Thiazole</td></tr></table>	Iupac name	Chemical class	octan-1-ol	Alcohol	cyclohex-3-ene-1-carbaldehyde	Aldehyde	heptanal	Aldehyde	hexanal	Aldehyde	nonanal	Aldehyde	octanal	Aldehyde	pentanal	Aldehyde	undecanal	Aldehyde	4,6-dimethyldodecane	Alkane	3,3-dimethyl-2-benzofuran-1-one	Furanone	6-methylhept-5-en-2-one	Ketone	1,3-benzothiazole	Thiazole	-) Aldehydes can originate from lipid autoxidation and beta-oxidation (Belitz et al., 2009) -) Heptanal, octanal, and nonanal can originate from Oleic acid (18:1) -) Pentanal, hexanal, and octanal can originate from linoleic acid (18:2) -) Fatty acid composition and consumption during incubation can depend on the flock ages of the mother hens (Sahan et al., 2014) -) 6-methylhept-5-en-2-one can originate from lycopene (carotenoid)														
Iupac name	Chemical class																																									
octan-1-ol	Alcohol																																									
cyclohex-3-ene-1-carbaldehyde	Aldehyde																																									
heptanal	Aldehyde																																									
hexanal	Aldehyde																																									
nonanal	Aldehyde																																									
octanal	Aldehyde																																									
pentanal	Aldehyde																																									
undecanal	Aldehyde																																									
4,6-dimethyldodecane	Alkane																																									
3,3-dimethyl-2-benzofuran-1-one	Furanone																																									
6-methylhept-5-en-2-one	Ketone																																									
1,3-benzothiazole	Thiazole																																									
Alcohol & ketones cluster  day	<table><tr><th>Iupac name</th><th>Chemical class</th></tr><tr><td>(1R,5R)-3,3,5-trimethylcyclohexan-1-ol</td><td>Alcohol</td></tr><tr><td>(6S)-6-methyloctan-1-ol</td><td>Alcohol</td></tr><tr><td>1-(2-methoxypropoxy)propan-2-ol</td><td>Alcohol</td></tr><tr><td>2-(2-butoxyethoxy)ethanol</td><td>Alcohol</td></tr><tr><td>2-phenylpropan-2-ol</td><td>Alcohol</td></tr><tr><td>dodecan-1-ol</td><td>Alcohol</td></tr><tr><td>nonan-1-ol</td><td>Alcohol</td></tr><tr><td>propane-1,2-diol</td><td>Alcohol</td></tr><tr><td>benzaldehyde</td><td>Aldehyde</td></tr><tr><td>dodecanal</td><td>Aldehyde</td></tr><tr><td>5-methyl-2,3-dihydrofuran</td><td>Furan</td></tr><tr><td>(5E)-6,10-dimethylundeca-5,9-dien-2-one</td><td>Ketone</td></tr><tr><td>1-(cyclohexen-1-yl)ethanone</td><td>Ketone</td></tr><tr><td>1-phenylethanone</td><td>Ketone</td></tr><tr><td>2,6-di-tert-butylcyclohexa-2,5-diene-1,4-dione</td><td>Ketone</td></tr><tr><td>3-methylcyclohex-2-en-1-one</td><td>Ketone</td></tr><tr><td>diphenylmethanone</td><td>Ketone</td></tr><tr><td>phenol</td><td>Phenol</td></tr><tr><td>diethyl benzene-1,2-dicarboxylate</td><td>Phthalate ester</td></tr></table>	Iupac name	Chemical class	(1R,5R)-3,3,5-trimethylcyclohexan-1-ol	Alcohol	(6S)-6-methyloctan-1-ol	Alcohol	1-(2-methoxypropoxy)propan-2-ol	Alcohol	2-(2-butoxyethoxy)ethanol	Alcohol	2-phenylpropan-2-ol	Alcohol	dodecan-1-ol	Alcohol	nonan-1-ol	Alcohol	propane-1,2-diol	Alcohol	benzaldehyde	Aldehyde	dodecanal	Aldehyde	5-methyl-2,3-dihydrofuran	Furan	(5E)-6,10-dimethylundeca-5,9-dien-2-one	Ketone	1-(cyclohexen-1-yl)ethanone	Ketone	1-phenylethanone	Ketone	2,6-di-tert-butylcyclohexa-2,5-diene-1,4-dione	Ketone	3-methylcyclohex-2-en-1-one	Ketone	diphenylmethanone	Ketone	phenol	Phenol	diethyl benzene-1,2-dicarboxylate	Phthalate ester	-) Reduction of aldehydes by alcohol dehydrogenase into alcohols (the enzyme activity can be increased by higher estradiol concentrations; Hiremath et al., 1992) -) Benzaldehyde, Phenol, and 1-phenylethanone have been identified in the past as sex-specific compounds (Webster et al., 2015) → This cluster might contain relevant VOCs toward sex differences
Iupac name	Chemical class																																									
(1R,5R)-3,3,5-trimethylcyclohexan-1-ol	Alcohol																																									
(6S)-6-methyloctan-1-ol	Alcohol																																									
1-(2-methoxypropoxy)propan-2-ol	Alcohol																																									
2-(2-butoxyethoxy)ethanol	Alcohol																																									
2-phenylpropan-2-ol	Alcohol																																									
dodecan-1-ol	Alcohol																																									
nonan-1-ol	Alcohol																																									
propane-1,2-diol	Alcohol																																									
benzaldehyde	Aldehyde																																									
dodecanal	Aldehyde																																									
5-methyl-2,3-dihydrofuran	Furan																																									
(5E)-6,10-dimethylundeca-5,9-dien-2-one	Ketone																																									
1-(cyclohexen-1-yl)ethanone	Ketone																																									
1-phenylethanone	Ketone																																									
2,6-di-tert-butylcyclohexa-2,5-diene-1,4-dione	Ketone																																									
3-methylcyclohex-2-en-1-one	Ketone																																									
diphenylmethanone	Ketone																																									
phenol	Phenol																																									
diethyl benzene-1,2-dicarboxylate	Phthalate ester																																									
Alkanes cluster  day	<table><tr><th>Iupac name</th><th>Chemical class</th></tr><tr><td>2-ethylhexan-1-ol</td><td>Alcohol</td></tr><tr><td>(E)-non-2-enal</td><td>Aldehyde</td></tr><tr><td>(E)-oct-2-enal</td><td>Aldehyde</td></tr><tr><td>hexadecane</td><td>Alkane</td></tr><tr><td>hexane</td><td>Alkane</td></tr><tr><td>methylcyclopentane</td><td>Alkane</td></tr><tr><td>tetradecane</td><td>Alkane</td></tr><tr><td>1-methylcyclopentene</td><td>Alkene</td></tr></table>	Iupac name	Chemical class	2-ethylhexan-1-ol	Alcohol	(E)-non-2-enal	Aldehyde	(E)-oct-2-enal	Aldehyde	hexadecane	Alkane	hexane	Alkane	methylcyclopentane	Alkane	tetradecane	Alkane	1-methylcyclopentene	Alkene	Fatty acid degradation (Belitz et al., 2009) -) Primary products autoxidation: (E)-non-2-enal and (E)-oct-2-enal from linoleic acid																						
Iupac name	Chemical class																																									
2-ethylhexan-1-ol	Alcohol																																									
(E)-non-2-enal	Aldehyde																																									
(E)-oct-2-enal	Aldehyde																																									
hexadecane	Alkane																																									
hexane	Alkane																																									
methylcyclopentane	Alkane																																									
tetradecane	Alkane																																									
1-methylcyclopentene	Alkene																																									



Finally, the aromatics cluster tends to have its highest proportion on day 0 with a subsequent decrease over time. This suggests the presence of potentially xenobiotic (i.e., foreign) compounds that were absorbed by the egg during or after laying and were subsequently re-emitted or metabolized during incubation. The clear distinction over time with the specific clustering of chemical classes demonstrates the potential for linking and interpreting chemical and biochemical events during embryonic development.

1.4) Comparison SIFT vs PTR

In section 1.2, the need was defined to look for a more sensitive instrument than the SIFT-MS such as a PTR-ToFMS. A collaboration was set up with the University of Antwerp (Belgium). We performed two comparison experiments on their PTR-ToFMS and our SIFT-MS. Hereto, measurements were performed on consumption eggs at room temperature due to practical reasons and the fact that consumption eggs do deliver a representative spectrum to indicate the suitability of the system for future experiments on hatching eggs. One experiment aimed for distinguishing eggs from different chicken management systems (Omega3-enriched, Free-range & Organic). The other experiment was to reveal differences in emitted volatiles by eggs stored at different temperatures for 6 days (at 7 °C, 25 °C & 35 °C). The instrument settings of both SIFT and PTR were set at equal or comparable parameters in terms of sampling flow and measurement time. The eggs (having 3 to 4 samples per group) were measured on the same day on both instruments under the same conditions.

Even though the SIFT-MS uses three reagent ions to increase its specificity, it was hypothesized that the PTR would compensate for this by having a higher resolution of the mass spectrometer and higher sensitivity, and a shorter sampling time. In the comparison experiment of management systems, it was possible to distinguish Organic eggs from Omega3-enriched and Free-range eggs using both SIFT and PTR. The cross-validations did not manage to distinguish the two latter groups from each other (Table 2).

Table 2: PLS-DA models on eggs from different management systems

	LV	AccCV1	AccCV2	AccCV3	RMSECV1	RMSECV2	RMSECV3	R ² 1	R ² 2	R ² 3
PTR	4	0.5000	0.7500	0.9000	0.7147	0.4693	0.4601	0.6712	0.0968	0.1281
SIFT	4	0.3000	0.8333	1.0000	0.4846	0.4930	0.3553	0.0193	0.0004	0.4857
PTR*	5	0.3000	0.8333	0.9000	0.7108	0.5886	0.2882	0.0006	0.0456	0.6565

*: 0.01 m/z PTR resolution data with a minimum value of 10 counts per product ion and median centering preprocessing of the reagent ion corrected data. group 1 (n = 3) "Omega3", group 2 (n = 2) = "FreeRange", group 3 (n = 3) = "Organic"

Regarding performance differences between the two instruments, no conclusive difference indicated the superiority of one system to another. Similar to the management system experiment with Organic eggs, the cold treatment could be separated from the other treatments in the storage temperature experiment using the PTR data. This was less pronounced in the SIFT data and the model parameters were more in favor of the PTR.

Table 3: PLS-models storage temperature PTR vs. SIFT

	LV	AccCV1	AccCV2	AccCV3	RMSECV1	RMSECV2	RMSECV3	R ² 1	R ² 2	R ² 3
PTR_20pct	5	1.0000	0.6875	0.7500	0.2635	0.4406	0.3643	0.7137	0.2250	0.4113
SIFT_20pct	7	0.9375	0.1875	0.6250	0.3871	0.6659	0.4775	0.3354	0.2441	0.0542

20 pct of the most abundant were used. Group 1 (n = 4) = "Cold", Group 2 (n = 4) = "Ambient", Group 3 (n = 4) = "Hot"

In conclusion, the PTR did not deliver the significant sensitivity improvement as expected. The results of both VOC analysis systems were comparable. Nevertheless, it is expected that the PTR might deliver complementary tools in terms of data preprocessing due to the higher resolution. Furthermore, the PTR is superior in terms of speed for acquainting full scan data. However, chances of success are expected to be higher when a white-box approach is applied. Therefore, the search was continued for a higher throughput extraction and measurement process on the GC-MS that is described in section 1.5.

1.5) Twister optimization

Challenges from the previous sections already indicated that eggs emit low VOC profiles and analysis instruments require to be highly sensitive. For these reasons, a new sensitive VOC measurement approach was defined in which a sufficient number of observations could be retrieved. This approach made use of PDMS Twister® stir bars as sorbents instead of the SPME fiber. These sorbents allowed to increase in the number of measurements for the white-box sexing experiments on the GC-MS.

Relative to the SPME fiber, the Twisters have a higher sorption capacity and have the possibility of running samples overnight since no person needs to be present. As a result, this increased the number of measurements from 8 per day with the SPME to 24 per day with the Twisters. Regarding sensitivity, a similar to better sensitivity was obtained using the Twisters. Furthermore, the SPME fiber sometimes had a higher selectivity for some compounds whereas the Twisters were covering a broader range. Regarding the GC-MS analysis, the program was shortened for the Twisters to optimize the number of samples measured per day. Due to the differences in inlet systems, its desorption program had to be optimized in preliminary experiments and is different than SPME. Regarding extraction, a Twister was put together with an egg in a closed jar for 2 hours (Figure 5). Thereafter, the Twister was loaded into a desorption tube that was positioned on the system awaiting its automatic desorption and subsequent analysis. Twisters remained stable for 24 hours up to a week after the extraction.

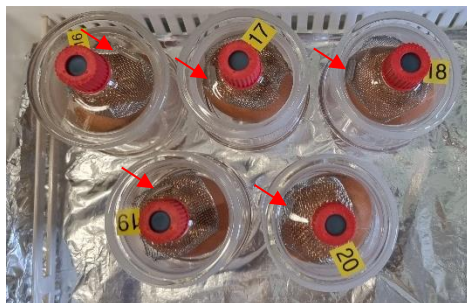


Figure 5: Eggs in glass jars incubating at 37.7 °C for 2 hours with a PDMS Twister (red arrows) on top of a stainless steel grid.

1.6) Twister fertile vs. infertile eggs

For this goal, 45 Isa Brown incubation eggs have been followed up for the even days of incubation from days 0 to 12. The measurement protocol described in the previous section was applied to extract VOCs from both fertilized and unfertilized eggs. From days 0 to 4, 16 females, 17 males, and 5 unfertilized eggs were daily measured. From days 6 to 12, more eggs were added and a total of 17 females, 17 males, and 7 unfertilized eggs were measured. The objective was to distinguish the egg compounds that were originating from biochemical processes and not only chemical processes. It was hypothesized that this knowledge would allow for a better distinction of sex-specific compounds that specifically originated from these biochemical processes.

A total of 114 compounds have been identified, this was almost double the number of compounds that have been identified in the experiment of section 1.3 using the SPME fiber (65 compounds). This confirmed the fact that Twisters had a broader range than the SPME. The identified compounds together with their chemical class are presented in Table 4. On the one hand, it was found that especially carboxylic acids were increased in their relative abundance for the fertilized eggs. On the other hand, the majority of compounds appeared to have relatively lower abundances in fertilized eggs. This was observed as well in the literature for compounds such as hexanal, decanal, and 1-phenylethanone (Webster, Hayes & Pike, 2015 & Xiang et al., 2018). It was hypothesized that, due to increased metabolism, there was an increase in carbon dioxide and a decrease in oxygen in the microenvironment of the egg. This might have subsequently led to oxidative degeneration of fatty- and amino acids into carboxylic acids, instead of aldehydes, alcohols, or ketones for instance (Figure 4).

Next, PLS-DA models were built to quantify the accuracy with which fertilized eggs could be distinguished from unfertilized eggs. As a result, it was observed that a 100 % prediction accuracy was obtained after 6 days of incubation. This was also the day when significant differences started to become visible in the abundance of carboxylic acids. Furthermore, the highest number of relevant VOCs for discriminating fertilized eggs was observed on day 10. In section 1.8, this day was also selected to build robust statistical models for *in ovo* sexing.

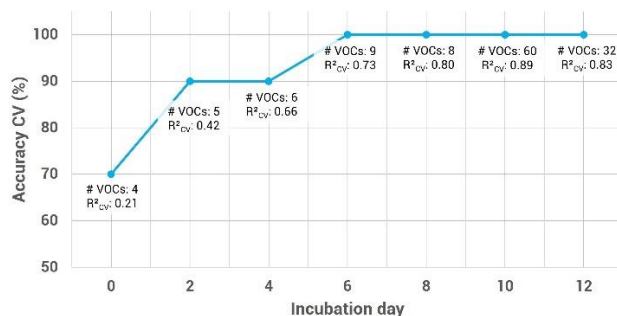


Figure 6: PLS-DA models on Fertilization status. The accuracy of the cross-validation (CV) is plotted versus the incubation day. On days 0/2/4, 5 fertilized eggs were compared to 5 unfertilized eggs. On days 6/8/10/12, 7 fertilized eggs were compared to 7 unfertilized eggs. A leave-one-out CV was performed using a Jackknifing variable selection. The number of retained variables in the model and the R^2 for the CV are depicted per day



Finally, t-tests were also performed on the individual compounds per day and are presented in Table 4. Next, in order to predict the sex, PLS-DA models were built similarly on the data of the 16 (or 17) females and 17 males measured per day (Figure 7). Since these datasets were limited, only the prediction accuracies on the cross-validation sets could be evaluated. As a result, it was observed that the prediction accuracies did not reach higher than 76 % (on day 4). It was concluded that none of the models succeeded to determine the sex on one of the days of incubation. Subsequently, it was hypothesized that a repeated experiment would deliver more robust and concluding results when performed on a larger dataset, using different VOC analysis systems (GC-, SIFT- & PTR-MS), on day 10 of incubation where the difference between fertilized eggs and unfertilized eggs was the largest. This study is discussed in section 1.8.

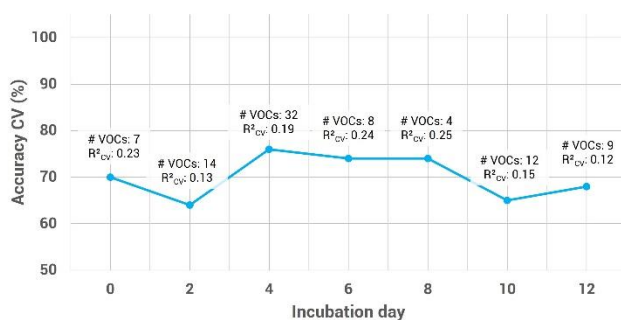


Figure 7: PLS-DA models on Fertilization status. The accuracy of the cross-validation (CV) is plotted versus the incubation day. On days 0/2/4, 16 female eggs were compared to 17 male eggs. On days 6/8/10/12, 17 female eggs were compared to 17 male eggs. A leave-one-out CV was performed using a Jackknifing variable selection. The number of retained variables in the model and the R^2 for the CV are depicted per day

Table 4: Identified egg volatile compounds on days 0/2/4/6/8/10/12 of incubation

IUPAC name	Chemical class	Significant days Fertilization (UF/Fe)	Significant days Sex (F/M)
1,2-diphenylethane-1,2-diol	Alcohol	10 (UF), 12 (UF)	0 (F), 4 (F)
dodecan-1-ol	Alcohol	10 (UF), 12 (UF)	
2-(2-hydroxypropoxy)propan-1-ol	Alcohol	8 (UF), 10 (UF), 12 (UF)	
2,4,7,9-tetramethyldec-5-yne-4,7-diol	Alcohol		
furan-2-ylmethanol	Alcohol	10 (UF), 12 (UF)	10 (F)
1-(2-hydroxypropoxy)propan-2-ol	Alcohol	8 (UF), 10 (UF), 12 (UF)	
hexadecan-3-ol	Alcohol	12 (UF)	4 (M)
2-phenylpropan-2-ol	Alcohol	0 (UF), 2 (UF), 8 (UF), 10 (UF), 12 (UF)	4 (F)
1-phenylethanol	Alcohol	8 (UF), 10 (UF), 12 (UF)	
phenylmethanol	Alcohol	8 (UF), 10 (UF), 12 (UF)	
benzene-1,2-diol	Alcohol	12 (UF)	10 (F)
2-(2-butoxyethoxy)ethanol	Alcohol	10 (UF), 12 (UF)	
2-dodecoxyethanol	Alcohol		
2-(2-hydroxyethoxy)ethanol	Alcohol	4 (UF), 10 (UF), 12 (UF)	
2-phenoxyethanol	Alcohol	10 (UF), 12 (UF)	
tridecan-1-ol	Alcohol	10 (UF), 12 (UF)	
1H-pyrrole-2-carbaldehyde	Aldehyde	10 (UF), 12 (UF)	
(Z)-dec-2-enal	Aldehyde		
(E)-oct-2-enal	Aldehyde		
(E)-undec-2-enal	Aldehyde		
benzaldehyde	Aldehyde	8 (UF), 10 (UF), 12 (UF)	
(E)-3-phenylprop-2-enal	Aldehyde	6 (Fe), 8 (Fe), 10 (Fe)	
decanal	Aldehyde	10 (UF), 12 (UF)	
Dodecanal	Aldehyde	10 (UF), 12 (UF)	
Heptanal	Aldehyde		2 (M)
hexanal	Aldehyde	6 (UF), 8 (UF), 10 (UF), 12 (UF)	
3-methylhexanal	Aldehyde	8 (UF), 10 (UF), 12 (UF)	
morpholine-4-carbaldehyde	Aldehyde		2 (F), 4 (F), 6 (F), 10 (F)
nonanal	Aldehyde	8 (UF), 10 (UF), 12 (UF)	
octanal	Aldehyde	8 (UF), 10 (UF), 12 (UF)	
pentadecanal	Aldehyde	12 (UF)	4 (M), 6 (M), 10 (M), 12 (M)
tetradecanal	Aldehyde	12 (UF)	2 (M), 4 (M), 6 (M), 10 (M), 12 (M)
tridecanal	Aldehyde	10 (UF), 12 (UF)	2 (M), 4 (M), 6 (M), 10 (M)
undecanal	Aldehyde	10 (UF), 12 (UF)	
4-hydroxy-3-methoxybenzaldehyde	Aldehyde		
1-iodododecane	Alkane	0 (UF), 4 (UF), 8 (UF), 10 (UF), 12 (UF)	
icosane	Alkane	4 (UF), 6 (UF), 8 (UF), 10 (UF), 12 (UF)	
heptadecane	Alkane	0 (UF), 2 (UF), 4 (UF), 10 (UF), 12 (UF)	
3-methylheptadecane	Alkane	4 (UF), 6 (UF), 10 (UF), 12 (UF)	
hexadecane	Alkane		2 (M)
octadecane	Alkane	4 (UF), 6 (UF), 8 (UF), 10 (UF), 12 (UF)	
tetracosane	Alkane	6 (UF), 10 (UF)	
N,N-dibutylformamide	Amide	8 (UF), 10 (UF), 12 (UF)	
hexanamide	Amide	12 (UF)	
3-methyl-3-phenylazetidine	Amine		
N,N-dimethyl-1-phenylmethanamine	Amine		
2-aminobenzonitrile	Amine		
hydrazine	Amine	2 (Fe), 6 (Fe)	
2,6-di(propan-2-yl)naphthalene	Aromatic		
9-methylidene fluorene	Aromatic	6 (UF), 8 (UF), 10 (UF), 12 (UF)	12 (M)
9H-xanthene	Aromatic	8 (UF), 10 (UF), 12 (UF)	
2-methylanthracene	Aromatic	6 (UF), 10 (UF), 12 (UF)	4 (M), 8 (M), 12 (M)
undecan-2-ylbenzene	Aromatic	4 (UF), 12 (UF)	
1-phenylmethoxynaphthalene	Aromatic	12 (UF)	
phenanthrene	Aromatic	6 (UF), 8 (UF), 10 (UF), 12 (UF)	12 (M)

Category abbreviations: Fe: Fertilized, UF: Unfertilized, F: Female, M: Male. The days that were significantly different based on a t-test are accompanied by the category for which a significantly higher value in relative abundance was observed.



Table 4 (continued): Identified egg volatile compounds on days 0/2/4/6/8/10/12 of incubation

IUPAC name	Chemical class	Significant days Fertilization (UF/Fe)	Significant days Sex (F/M)
4-phenylmethoxybenzoic acid	Carboxylic acid	6 (UF)	4 (M), 8 (M)
acetic acid	Carboxylic acid		
benzoic acid	Carboxylic acid		6 (F)
Benzoic acid, (4-benzoyloxy-2-chlorophenyl) ester	Carboxylic acid	12 (UF)	
butanoic acid	Carboxylic acid	6 (Fe), 8 (Fe), 10 (Fe), 12 (Fe)	
heptanoic acid	Carboxylic acid	6 (Fe), 8 (Fe), 10 (Fe), 12 (Fe)	
2-ethylhexanoic acid	Carboxylic acid	10 (Fe), 12 (Fe)	
decanoic acid	Carboxylic acid	8 (Fe), 10 (Fe), 12 (Fe)	
nonanoic acid	Carboxylic acid	6 (Fe), 8 (Fe), 10 (Fe), 12 (Fe)	
octadecanoic acid	Carboxylic acid	4 (Fe)	
octanoic acid	Carboxylic acid	6 (Fe), 8 (Fe), 10 (Fe), 12 (Fe)	
tetradecanoic acid	Carboxylic acid		
2,3-diacetyloxypropyl acetate	Carboxylic acid	8 (UF), 12 (UF)	
3H-2-benzofuran-1-one	Ester	10 (UF), 12 (UF)	
dimethyl benzene-1,4-dicarboxylate	Ester	8 (UF), 10 (UF), 12 (UF)	
1,6-dioxacyclodecane-7,12-dione	Ester	0 (UF), 12 (UF)	6 (F)
3-[2-(2-methoxyethoxy)ethoxy]prop-1-ene	Ester		
5-pentylloxolan-2-one	Ester	8 (UF)	
butyl (E)-but-2-enoate	Ester		
2-ethylhexyl 2-hydroxybenzoate	Ester	8 (UF), 12 (UF)	
(3,3,5-trimethylcyclohexyl) 2-methylprop-2-enoate	Ester		
2-phenoxyethoxybenzene	Ester	6 (UF), 8 (UF), 10 (UF), 12 (UF)	
methyl 2-(3-oxo-2-pentylcyclopentyl)acetate	Ester	12 (UF)	4 (M)
1-ethenoxydecane	Ester	12 (UF)	4 (M)
dodecyl prop-2-enoate	Ester	10 (UF), 12 (UF)	
phenacyl formate	Ester		
(2-acetyloxy-3-hydroxypropyl) acetate	Ester	8 (UF), 12 (UF)	
1,3-dihydroxypropan-2-yl hexadecanoate	Ester		
propan-2-yl tetradecanoate	Ester	12 (UF)	4 (M)
methyl 9-oxononanoate	Ester		6 (M), 10 (M), 12 (M)
1-octoyoctane	Ester	10 (UF), 12 (UF)	
p-Anisic acid, 4-nitrophenyl ester	Ester	12 (UF)	
Phthalic acid, di(2-propylpentyl) ester	Ester	10 (UF)	
Propanoic acid, 3-chloro-, 4-formylphenyl ester	Ester	6 (Fe), 8 (Fe), 10 (Fe), 12 (Fe)	
Terephthalic acid, isobutyl butyl ester	Ester	12 (UF)	
methyl (Z)-N-hydroxybenzenecarboximidate	Ether		
3-methyl-2H-furan-5-one	Furanone	10 (UF), 12 (UF)	4 (F), 8 (F), 10 (F)
1H-indole	Indole		
2,3-dihydroinden-1-one	Ketone	10 (UF), 12 (UF)	0 (F), 4 (F), 8 (F), 10 (F)
2-hydroxy-3-methylcyclopent-2-en-1-one	ketone	10 (UF), 12 (UF)	8 (F), 10 (F)
6,10,14-trimethylpentadecan-2-one	Ketone	4 (UF), 6 (UF), 8 (UF), 10 (UF), 12 (UF)	4 (M)
pyrrolidin-2-one	Ketone	10 (UF)	4 (F)
(5E)-6,10-dimethylundeca-5,9-dien-2-one	Ketone	8 (UF), 10 (UF), 12 (UF)	
fluoren-9-one	Ketone	6 (UF), 10 (UF), 12 (UF)	4 (M)
propan-2-one	Ketone		2 (F)
1-phenylethanone	Ketone	8 (UF), 10 (UF), 12 (UF)	2 (F), 4 (F)
diphenylmethanone	Ketone	6 (UF), 8 (UF), 10 (UF), 12 (UF)	
1-(2,3-dihydro-1H-inden-5-yl)ethanone	Ketone		
1-(4-acetylphenyl)ethanone	Ketone		
pyrrolidine-2,5-dione	Ketone	10 (UF)	4 (F)
2-phenylphenol	Phenol	10 (UF)	
phenol	Phenol	10 (UF), 12 (UF)	0 (F), 4 (F), 10 (F)
2,4-dichlorophenol	Phenol		
5-ethenyl-2-methoxyphenol	Phenol		8 (F)
tris(1-chloropropan-2-yl) phosphate	Phosphate		4 (M)
bis(2-methylpropyl) benzene-1,2-dicarboxylate	Phthalate ester	12 (UF)	4 (M)
2,7-dimethylquinoline	Quinoline		
methylsulfonylmethane	Sulfide		
sulfur dioxide	Sulfide	10 (UF), 12 (UF)	0 (F), 4 (F), 8 (F)

Category abbreviations: Fe: Fertilized, UF: Unfertilized, F: Female, M: Male. The days that were significantly different based on a t-test are accompanied by the category for which a significantly higher value in relative abundance was observed.



1.7) Incubation stages discrimination SIFT vs. PTR

This section served the purpose of obtaining conclusive results on the most optimal VOC analysis system for hatching egg volatiles. More specifically, a comparison experiment was set up to compare the SIFT-MS relative to the PTR-ToFMS, with the Twister-GC-MS as a standard method. Therefore, 15 eggs from 3 different breeds were measured on days 2 and 8 of incubation on both three instruments. During the measurements, the eggs were individually placed in FEP bags together with a Twister in a grid holder (Figure 2c).

Due to the fact that the PTR-ToFMS had been in repair for a long period, these measurements have only taken place in the last month of the project. Subsequently, these data have not been fully analyzed yet and the time was used to finalize the analysis of the day 10 sexing experiment on the PTR-ToFMS that has taken place in the same period. Since this final sexing experiment was situated in the critical path of the project, this was prioritized first. Nevertheless, the data of this section will still be analyzed for writing an academic paper given its novelty and high relevance in defining the potential for other agricultural applications as well.

1.8) Sex-specific volatiles identification using Twister/SIFT/PTR

In this final section, results are discussed of 100 eggs measured on incubation day 10 on both the SIFT-MS, PTR-ToFMS, and Twister-GC-MS to verify the robustness of PLS-DA models for *in ovo* sexing. The data were obtained in two separate experiments, namely 1 experiment on the SIFT-MS and another experiment on both the PTR-ToFMS and Twister-GC-MS. For these experiments, eggs were individually incubated in FEP bags during an incubation time of 2 hours at 37.7 °C before the headspace was analyzed on the SIFT- or PTR-MS or when the Twister was removed from the bag. Since both three systems generate numerous variables and risk having the effect of random noise fitting, it was important to verify the robustness of the results. Therefore, the main objective of this study was to split the datasets into a 60 % calibration set and a 40 % validation set to test the reliability of the PLS-DA models in identifying embryonic sex.

Different PLS-DA models have been built on the three datasets. For the GC-MS data, the same 114 variables were identified from the library obtained on the dataset in section 1.6. As a result, a prediction accuracy of 81.67 % was obtained on the cross-validation set. However, only an accuracy of 55 % was obtained on the validation set. The same pattern was observed in the data of the SIFT-MS and the PTR-ToFMS, even for the most optimal models. For the SIFT-MS, 100 % prediction accuracy was obtained on the cross-validation set, whereas only a 60 % was obtained on the validation set. When concerning the PTR-ToFMS data, different preprocessing techniques were applied to the continuous spectra to improve the data for model building. The most optimal model was obtained after performing a standard normal variate, followed by a second-order Savitzky-Golay smoothing and a mean centering. As a result, a prediction accuracy of 91.15 % was obtained on the cross-validation. However, applying the model to the validation set resulted in a 60.44 % accuracy.



It was concluded from this study that prediction models on the validation sets were not providing satisfying results towards *in ovo* sexing on a single day of incubation using big sample numbers (i.e., 100 samples on day 10 measured on both Twister-GC-MS, SIFT-MS, and PTR-ToFMS). Nevertheless, researchers indicated in the literature significant differences between certain VOCs and the sex of the embryo. Similarly, in the data of section 1.6, we also observed statistical differences between the embryos for specific VOCs on specific days (Table 4). Furthermore, it was also observed that if a specific VOC was significantly higher for one of the two sexes, this pattern was also conserved for this sex on the other days (Table 4). As an example, the dynamic patterns of phenol, 1-phenylethanol, tetradecanal, and pentadecanal are presented in Figure 8.

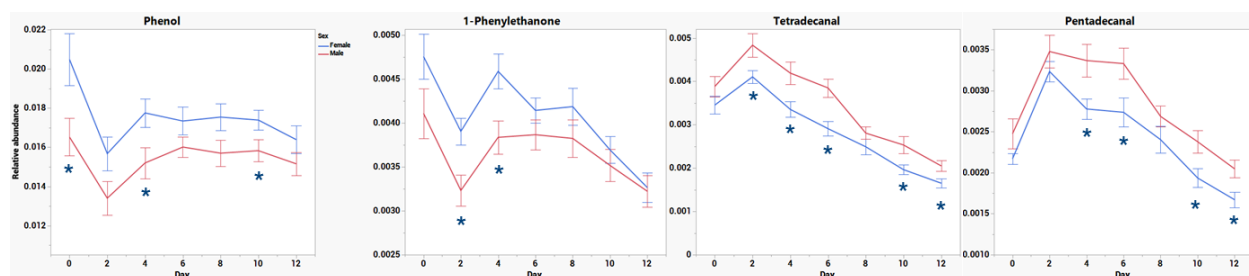


Figure 8: Average dynamic patterns for 16 females and 17 males of the relative abundances of 4 VOCs over 12 days of incubation. Significant differences are indicated with an asterisk ($P < 0.05$)

On the one hand, phenol (alcohol) and 1-phenylethanol (ketone) were higher for females throughout incubation. This higher abundance in females was also observed for quail eggs on day 8 (Webster, Hayes & Pike, 2015). On the other hand, tetradecanal and pentadecanal (aldehydes) were higher in their relative abundance for males. From the literature, it is known that the enzyme, alcohol dehydrogenase, reduces aldehydes into alcohols. Furthermore, it was demonstrated that the activity of alcohol dehydrogenase increased with higher levels of estradiol (Hiremath et al., 1992). Since estradiol is a female hormone, it was hypothesized that potentially increased enzyme activity due to higher estradiol concentrations in females might have resulted in a higher reduction of aldehydes into alcohols in females and thus eventually a lower production of these aldehydes in females.

From a future perspective, even though we did not manage to build a successful sexing model on a single incubation day (see section 1.6 on a limited dataset and section 1.8 on a large dataset). It can be considered to take into account the dynamic pattern of specific VOCs to improve the performance of the model. Namely, by using this longitudinal information, it is assumed that the differences between the sexes might be more distinct. Practically, this would mean that eggs have to be monitored at different moments in incubation. However, in industrial implementations, these extra manipulations would make the process more complex.

2.0) & 2.1) Glue selection & adsorbent identification setup preparation & Construction of volatile accumulator

This section contains the preliminary results on the construction of the volatile accumulator and the adsorbent selection. First, an adhesive selection was performed based on different characteristics such as dry-out speed, odor, flexibility, spread, and cost. This adhesive was required to adhere the volatile accumulator airtight on the eggshell without producing too many extra odors. A total of five different silicone- or acrylate-based adhesives were tested. A silyl-modified polymer-based adhesive, Tec7, was selected to be used for the accumulator and was also used to assemble the flow-through chips later on.

Second, preliminary steps were taken for the adsorbent selection. Here, a linear flow-through microfluidic chip of six different dyes was built and consisted of gas-responsive dyes such as Fluorescein/ Phenol red/ Reichardt's dye/ Bromophenol blue/ Methylene blue and Acridine orange (Figure 9). These dyes were deposited in an ormosil solution on different substrates, such as PVDF, cotton, and silicone-impregnated paper. A flow-through of certain gases at a known concentration was generated with a peristaltic pump and color changings were measured by positioning the chip inside a tabletop flatbed scanner. In this phase, it was more consistent and user-friendly by using a scanner rather than a hyperspectral camera to quantify the color shifts.

The most important findings were made on the substrate selection, the formulation protocol, and the dye selection. First, it was concluded that a PVDF membrane was the preferred substrate for the ormosil with dye deposition since there was a better optimum between pre-concentration and surface activity resulting in a higher signal. Second, nitrogen curing improved the sensitivity. This might be explained by the fact that oxidation and other irreversible reactions with air compounds were avoided. Third, R-dye/ Fluorescein and methylene blue were the most responsive dyes. However, they did not react visibly to gas concentrations at 100 ppm of for instance benzaldehyde. As a solution, it was proposed to use porphyrins for increasing sensitivity and specificity. These porphyrins would have been the next step in improving the setup. However, the scope was then shifted to first identifying the sex-specific gases using more sensitive techniques such as GC-, SIFT, and PTR-MS

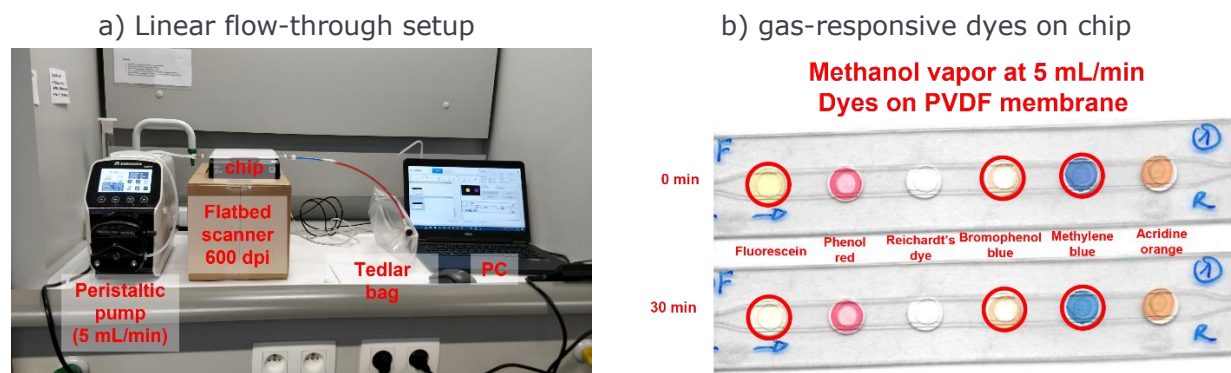


Figure 9: Setup for sensitivity measurements of dyes relative to the flow through of specific gases. a) Picture of the setup. b) Gas-responsive dyes in which the changing dyes have been indicated



References:

- Belitz, H.-D., W. Grosch, and P. Schieberle. 2009. Food Chemistry. 4th revise.
- Hiremath, L. S., P. M. Kessler, G. C. Sasaki, and P. E. Kollattukudy. 1992. Estrogen induction of alcohol dehydrogenase in the uropygial gland of mallard ducks. Eur. J. Biochem. 203:449–457.
- Webster, B., W. Hayes, and T. W. Pike. 2015. Avian egg odour encodes information on embryo sex, fertility and development. PLoS One 10 Available at <http://dx.doi.org/10.1371/journal.pone.0116345>.
- Xiang, X. le, Y. lan Wang, Z. hui Yu, M. hu Ma, Z. hui Zhu, and Y. guo Jin. 2018. Non-destructive characterization of egg odor and fertilization status by SPME/GC-MS coupled with electronic nose. J. Sci. Food Agric. 99:3264–3275.

2.7. Discuss efforts taken to ensure the approach is scientifically rigorous. Include approaches taken to ensure robust and unbiased results.

Throughout the whole project, it was observed that egg VOC measurements required low background signal and a high number of measurements to be able to build robust statistical models.

The background noise was lowered by using inert materials such as glass jars and FEP Teflon bags. These materials were also cleaned using filtered air, heated air purging, or baking out in an oven.

Next to that, the risk was existing of random noise fitting which happens when you have a low number of observations and a high number of variables. Subsequently, it is possible that you randomly fit a good model on the data, although this might not be reproducible on an independent test set. Moreover, VOC screening measurements are a lengthy process and limit the number of observations. To cope with this, a new extraction approach was developed using PDMS Twisters. This allowed us to sample 100 eggs on a single day and analyze them afterward on a GC-MS instrument. Furthermore, the sample size was also increased to 100 eggs on both the SIFT-MS and PTR-ToFMS to obtain representative datasets for building reliable models.



2.8. List key stakeholders that could be served by results of this project.

Concerning the results of this project, different key players in the poultry industry would be interested to know about the outcome. First, egg grading companies have interest since they will be the first ones to adapt their systems for *in ovo* sexing applications. Examples are Viscon, Moba, Ceva, Ovotrack, and Innovatec. Second, incubator manufacturers also closely follow up on the latest developments in the field. Examples are Petersime, Hatch tech, and Pasreform. Finally, genetic breeding companies will also be interested since it concerns innovative research on the eggs of their genetic lines. Examples are Hendrix Genetics, Lohmann Tierzucht, Aviagen, and Cobb-Vantress.

2.9. How have the results of this reporting period been disseminated to communities of interest? Describe how the results have been disseminated to communities of interest. Include any outreach activities that have been undertaken to reach members of communities who are not usually aware of such activities, to enhance public understanding and increase interest in learning and careers in science, technology, and the humanities. Reporting the routine dissemination of information (e.g., websites, press releases) is not required. For awards not designed to disseminate information to the public or conduct similar outreach activities, a response is not required; the grantee should write "nothing to report." A detailed response is only required for awards or award components that are designed to disseminate information to the public or conduct similar outreach activities. Note that scientific publications and sharing research sources will be reported under Information Products. (up to 1,000-word limit)

Similar to last year, the research outcome was presented at a conference (this time in real life in Leiden, The Netherlands). This conference was organized by the Incubation & Fertility Research Group (IFRG) on the 13-14th of October 2022. This research group works specifically on avian embryonic development and is the most relevant research group of the world poultry science association to present the results. The work was presented during an oral presentation of 15 minutes followed by a 5 minutes questions and answers session. As feedback, the experts were impressed by the presented work on differentiating fertilized from unfertilized eggs and the sex determination of embryos based on VOCs during the first 12 incubation days. Finally, the Nick French Award for the best presentation by a young scientist was assigned to our work for a second time (<https://ifrg.be/>).



2.10. Describe challenges or delays encountered during project and actions that weretaken to resolve them. Only describe significant challenges that may have impeded the research and emphasize their resolution. (up to 500-word limit)

The following challenges have also been described in the previous reports of the grant and the major ones are listed here.

- 1) Generally, the sensitivity and the number of measured samples in function of time were one of the most important challenges. This forced us to search for novel measurement systems to capture low-abundant VOCs and to optimize and validate them. For example, PDMS Twisters have been used to deliver the most optimal combination of the highest sensitivity and sample throughput on a GC-MS instrument. However, given its novelty, substantial time was required to optimize the setup.
- 2) Given the low abundance of egg signals it is important to lower the background signals as much as possible and prevent carry-over from one sample to another. This risk was reduced at the level of the GC-MS by adding an extra cleaning step for the Twisters using a tube conditioner module. At the level of the SIFT and PTR, a thermal bag cleaner module was used to flush the bag containing an egg prior to headspace incubation. An ozone module was also used to clean empty bags. However, it was observed that this cleaning was relatively aggressive and it even increased the background signal by releasing compounds from the bag material.
- 3) The preliminary experiments on the design of the chemical sensor indicated that this VOC analysis technique is less sensitive than a GC-, PTR-, or SIFT-MS. Therefore, its development was set on hold to focus on the specific identification of the biomarker. Using this biomarker, a more focused adsorbent and dye could be selected for the sexing. However, within the time frame of the project, it was not possible to define a single compound responsible for sex differentiation between days 0 to 12 of incubation. Subsequently, the development of this chemical sensor was not continued.
- 4) Finally, we faced a general backlog due to the COVID-19 lockdown in 2020. Given this fact we requested and received a no-cost extension of 6 months.



2.11. What opportunities for training and professional development has the project provided during this reporting period? If the research is not intended to provide training and professional development during this period, state "Nothing to Report." For all projects reporting graduate student and/or post-doctoral participants, grantees are encouraged to describe how Individual Development Plans (IDPs) are used to help manage the training for those individuals. (up to 500-word limit)

The research is part of the professional development of graduate student Matthias Corion. His individual development plan is followed-up and supervised by his main promotor and PI Prof. Jeroen Lammertyn. The development goals and milestones are in accordance with the Arenberg Doctoral School.

2.12. Please indicate the number of undergraduate and graduate students, post-doctoral scholars, or other educational components involved during this reporting period. If other education components are involved, please describe them in detail. (up to 500-word limit)

Similar to the previous year, the educational "components" involved in this reporting period were an undergraduate student who is working on his master thesis together with one graduate student who performs his Ph.D. on this topic.

3. Information Products

3.1. Please list the type(s) of information products (e.g., scholarly publications, reports or monographs, workshop summaries or conference proceedings, video, audio, images, models, software, curricula, instruments or equipment, intervention, etc.) produced during the project resulting directly from the FFAR award.

Two conference proceedings with an oral presentation in the IFRG meeting on the 14th of October 2022 entitled:

"In ovo sexing technologies for the poultry industry: insights from 1907 until 2021" (the scientific review paper on this will soon be submitted for publication)

"Determining the fertility and the sex of chicken hatching eggs using a new volatile organic compound extraction approach"

3.2. Please provide a list of citations for the information products produced during the project.

Nothing to report



- 3.3. Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during the project resulting directly from the FFAR award? If yes, please provide citation.**

Nothing to report

- 3.4. Website(s). List the URL for any internet site(s) that disseminates the research activities. A short description of each site should be provided. It is not necessary to include the publications already specified above.**

<https://www.bi.w.kuleuven.be/biosyst/mebios/biosensors-group>: The main MeBioS-Biosensors group page on which the disseminated research activities are presented of the lab.

- 3.5. Have inventions, patent applications, and/or licenses resulted from the project (U.S. and international)? If yes, indicate the invention, patent application(s), and/or license(s).**

	# Filed (Enter Numeric Value)	# Approved (Enter Numeric Value)	Patent Numbers (Separated by commas)
Patents	EP20194091.3 (collection gas chamber) EP20194084.8 (chemical sensor chip)	These applications have been retracted before the end of the priority year without outstanding rights since it would be too early to disclose the concept and further research was required first.	
Copyrights	Nothing to report	#	
Trademarks	Nothing to report	#	
Inventions	Nothing to report	#	



**FOUNDATION FOR
FOOD & AGRICULTURE
RESEARCH**

3.6. Are any of the information products produced during this grant that are confidential, proprietary, or subject to special license agreements? If so, please list them below and describe why they must remain confidential. Also, note if (and when) you plan to make these data publicly available in the future or if they must remain confidential indefinitely. (up to 500-word limit)

Yes, the goal is to valorize the performed research and results in a proof of concept for an industrial application. The IP will be protected by a patent.

The approach we follow towards *in ovo* sexing is innovative and requires careful consideration to maximize later impact. We are in close contact with our tech Transfer Office to investigate the optimal routes to IP protection. It is believed that these paths can form the subject of patent applications. In order to enable patenting, the developed solution will be kept confidential until a patent application is filed.

3.7. Beyond depositing information products in a repository, what other activities have you undertaken to ensure that others (e.g., researchers, decision makers, and the public) can easily discover and access the listed information products? What other activities have you undertaken to ensure that other can access and re-use these data in the future?

The general insights will be published in accordance with the KU Leuven Open Access (OA) policy. More specifically, Green OA is supported together with the deposit obligation in the institutional repository Lirias. Furthermore, Fair OA or Gold OA, is also supported.



4. Data Management

4.1. Did the project generate any data? Data generation includes transformation of existing data sets and data from existing resources (e.g., maps and imagery). Please list the data generated for the award.

Yes, data was generated on the SIFT-MS, PTR-ToFMS, and GC-MS

4.2. If you list multiple data sets, are these data sets related? If so, please provide a short description of how they are related.

Yes, datasets are related. In multiple experiments, eggs were both measured on the GC-MS, the SIFT-MS, or the PTR-ToFMS.

4.3. Please provide copies of relevant metadata records to support FFAR's mission of enhancing the discoverability of FFAR funded project data and information products. Include or attach copies of records and a simple file inventory, if necessary, in a compressed folder.

- **Exp 1 Fiber-GCMS optimization:** data & analysis files on table eggs
- **Exp 2 SIFT-MS FEP bags optimization:** data & analysis files on table eggs
- **Exp 3 Optimization incubation time Fiber-GCMS:** data & analysis files on table eggs
- **Exp 4 Preliminary Fiber-GCMS & SIFT-MS days 0 to 12:** data & analysis files on hatching eggs
- **Exp 5 Fiber-GCMS sexing days 0 & 10:** data & analysis files on hatching eggs
- **Exp 6 Fiber-GCMS & SIFT days 0 to 12:** data & analysis files on 7 hatching eggs measured per day
- **Exp 7 SIFT optimizations & Twisters vs fiber comparison:** data & analysis files on table eggs
- **Exp 8 SIFT vs PTR-ToFMS:** data & analysis files on table eggs
- **Exp 9 Twister-GCMS optimizations:** data & analysis files on table eggs
- **Exp 10 Twister-GCMS preliminary days 0 to 12:** data & analysis files on hatching eggs
- **Exp 11 SIFT egg headspace accumulation tests:** data & analysis files on table eggs
- **Exp 12 Twister-GCMS sexing days 8-10-12:** data & analysis files on 50 hatching eggs
- **Exp 13 Twister & SIFT comparison - target scans & bag cleaning:** data & analysis files
- **Exp 14 Twister-GCMS shortening & software comparison:** data & analysis files on table eggs
- **Exp 15 Twister-GCMS days 0-12:** data & analysis files for 45 hatching eggs for days 0 to 12
- **Exp 16 PTR vs. SIFT vs. Twister-GCMS:** data & analysis files of 15 hatching eggs for 3 different breeds
- **Exp 17 SIFT sexing D10:** data & analysis files of 100 hatching eggs
- **Exp 18 PTR & Twister-GCMS sexing D10:** data and analysis files of hatching 100 eggs
- **Experiments on adhesive & adsorbent selection:** pictures of colorimetric chips & analysis files



Certification

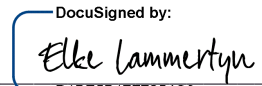
The undersigned hereby certifies that to the best of their knowledge and belief, the above report and all supporting documents are true, accurate and complete. I am aware there is a significant penalty for submitting false or misleading information.

If applicable, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I will not claim otherwise.

PI Full Name: Dr. Jeroen Lammertyn

PI Signature:  Date: 27 oktober 2022 | 05:58 PDT

Authorized Signing Official (ASO) Name: Ms. Elke Lammertyn

ASO Signature:  Date: 27 oktober 2022 | 14:59 CEST